

$\frac{1}{2}$

$$I = \int_0^{\pi} \frac{1}{1+m^2 \sin^2 \theta} d\theta$$

$$\forall m \in [-1, 1], \sum_{k=0}^{\infty} (-m^2)^k = \sum_{k=0}^{\infty} (-m^2)^k = \frac{1 - (-m^2)^{\infty}}{1 - (-m^2)} = \frac{1}{1+m^2}$$

$n \rightarrow \infty$

pour $m = -1$ ou 1

$$\sum_{k=0}^{\infty} (-m^2)^k = \sum_{k=0}^{\infty} (-1)^k = 1 + \sum_{k=1}^{\infty} (-1)^k = 1 + \sum_{k=1}^{\infty} (-1)^{k-1} = 1 - \sum_{k=0}^{\infty} (-1)^k = 1 - \sum_{k=0}^{\infty} (-1)^k$$

Donc $\sum_{k=0}^{\infty} (-m^2)^k = \frac{1}{2} = \frac{1}{1+m^2}$

Donc $\forall m \in [-1, 1], \sum_{k=0}^{\infty} (-m^2)^k = \frac{1}{1+m^2}$

car $\forall m \in \mathbb{R}$ comme $|m| \in [-1, 1], \frac{1}{1+m^2} = \sum_{k=0}^{\infty} (-m^2)^k$

donc $I = \int_0^{\pi} \sum_{k=0}^{\infty} (-m^2 \sin^2 \theta)^k d\theta$

$$= \sum_{k=0}^{\infty} (-1)^k \int_0^{\pi} \sin^{2k} \theta d\theta$$

ou $J_k(\theta) = \int_0^{\pi} \sin^{2k} \theta d\theta$

donc $J_k(\theta) = \int_0^{\pi} \sin^{2k} \theta d\theta$

$$= \int_0^{\pi} \frac{1}{2^k} \sum_{j=0}^k C_{2k}^j (\cos \theta)^j \sin^{2k-j} \theta d\theta$$

$$= \frac{1}{2^k} \sum_{j=0}^k C_{2k}^j \sin^j \theta \int_0^{\pi} \cos^{2k-j} \theta d\theta + \frac{1}{4^k} (C_{2k}^k \sin^k \theta \int_0^{\pi} \cos^k \theta d\theta) \quad j \neq k$$

$$= \frac{1}{4^k} \sum_{j=0}^k C_{2k}^j \sin^j \theta \left(\int_0^{\pi} \cos^{2(j-k)} \theta d\theta - \int_0^{\pi} \cos^{2(j-k)} \theta d\theta \right) + \frac{1}{4^k} (C_{2k}^k \sin^k \theta \int_0^{\pi} \cos^k \theta d\theta)$$

$$= \frac{1}{4^k} \left(\frac{\pi}{2} - \frac{2}{2(j-k)} \left(\int_0^{\pi} \cos^{2(j-k)} \theta d\theta - \int_0^{\pi} \cos^{2(j-k)} \theta d\theta \right) \right) + \frac{1}{4^k} C_{2k}^k \sin^k \theta \int_0^{\pi} \cos^k \theta d\theta$$

$$= \frac{1}{4^k} \sum_{j=0}^k C_{2k}^j \sin^j \theta \frac{\pi (j-k-1)}{-2(j-k)^2} + \frac{C_{2k}^k \pi^3}{3 \times 4^k} \quad j \neq k$$

Donc $J_k(\theta) = \frac{1}{4^k} \left(\sum_{j=0}^{2k} C_{2k}^j \sin^j \theta \frac{(\sin(j-k)-1)}{(j-k)^2} + \frac{C_{2k}^k \pi^3}{3} \right)$

Donc $J_k(\theta) = \frac{1}{4^k} \left(\frac{\pi^3}{3} - \sum_{j=0}^{2k} C_{2k}^j \sin^j \theta \frac{(\sin(j-k)-1)}{(j-k)^2} \right) \quad j \neq k$

$$\text{donc } I = \sum_{k=0}^{+\infty} \frac{(-1)^k}{4^k} \left(\frac{1^2 C_{2k}^k}{3} - \frac{1}{2} \sum_{j=0}^{2k} C_{2k}^j (-1)^{j+k} \frac{(\ln(j-k)-1)}{(j-k)^2} \right) \quad j \neq k$$

$$\text{Donc } I = \int_0^\pi \frac{x^2}{1+x^2} dx = \pi \sum_{k=0}^{+\infty} \frac{1}{4^k} \left(\frac{1^2 C_{2k}^k}{3} - \frac{1}{2} \sum_{j=0}^{2k} C_{2k}^j (-1)^j \frac{(\ln(j-k)-1)}{(j-k)^2} \right) \quad j \neq k$$

ou encore en remarquant que

$$\int \frac{x^2}{1+x^2} dx = \frac{1}{4\sqrt{2}} i \left(-2i\pi \operatorname{Li}_2(1-2\sqrt{2})e^{2ix} \right) + 2i\pi \operatorname{Li}_2(1+2\sqrt{2})e^{2ix} + \operatorname{Li}_2(1-2\sqrt{2})e^{2ix} - \operatorname{Li}_2(1+2\sqrt{2})e^{2ix} + 2i\pi \log(1+(2\sqrt{2}-3)e^{2ix}) - 2i\pi \log(1-(3+2\sqrt{2})e^{2ix})$$

$$I = \frac{1}{4\sqrt{2}} i \left(-2i\pi \operatorname{Li}_2(1-2\sqrt{2}) + 2i\pi \operatorname{Li}_2(1+2\sqrt{2}) + \operatorname{Li}_2(1-2\sqrt{2}) - \operatorname{Li}_2(1+2\sqrt{2}) + 2i\pi \log(-2+2\sqrt{2}) - 2i\pi \log(-2-2\sqrt{2}) + 0 - 0 - \operatorname{Li}_2(1-2\sqrt{2}) + \operatorname{Li}_2(1+2\sqrt{2}) \right)$$

$$\text{Donc } I = \frac{\pi\sqrt{2}}{4} \left(\operatorname{Li}_2(1-2\sqrt{2}) - \operatorname{Li}_2(1+2\sqrt{2}) + i\pi \log\left(\frac{2\sqrt{2}-1}{1-2\sqrt{2}}\right) \right) \\ = \frac{\pi\sqrt{2}}{4} \left(\operatorname{Li}_2(1-2\sqrt{2}) - \operatorname{Li}_2(1+2\sqrt{2}) + i\pi \log(e^{i\pi} \times (1-2\sqrt{2})) \right) \\ = \frac{\pi\sqrt{2}}{4} \left(\operatorname{Li}_2(1-2\sqrt{2}) - \operatorname{Li}_2(1+2\sqrt{2}) + \pi + i\pi \log(1-2\sqrt{2}) \right)$$

ou encore simplement

$$I = \lim_{n \rightarrow +\infty} \frac{\pi}{n} \sum_{k=1}^n \frac{\left(\frac{\pi k}{n}\right)^2}{1 + \tan^2 \frac{\pi k}{n}} = \lim_{n \rightarrow +\infty} \frac{\pi^3}{n^3} \sum_{k=1}^n \frac{k^2}{1 + \tan^2 \frac{\pi k}{n}}$$